

WILL MCGOWAN

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EDUCATION

University of Strathclyde, Glasgow

Sep 2023 – Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1, including 90% in first-year Finance
- Relevant Modules:** Applied Econometrics, Treasury Management and Derivatives, Advanced Microeconomics

Dissertation | MATLAB

In progress, due 2027

"Who Hedges the Barrel? Sovereign Oil Hedging and Petro-Currency Reactions to OPEC News"

- Testing three predictions fixed in advance: sovereign hedges should reshape oil betas like options, smaller under Norway's krone conversions, kinked at Russia's budget-rule cut-off, one-sided under Mexico's puts
- Measuring each currency's move on OPEC announcement days since 1983, net of the dollar's own move, with unhedged Canada as a control and difference-in-differences around Russia's rule changes
- Testing a pre-registered trading rule on any residual beta after costs, with a White reality check and Kelly sizing

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

PROJECTS

Plains Heat and the KC-Chicago Wheat Spread

R

- Tested whether Plains heatwaves move Kansas City wheat against Chicago, using the spread so shared news cancels, 27 summers of NOAA data and the off-season as a placebo: heat explained 0.3% of weekly moves
- Distrusted a too-good lead-lag signal (0.74 correlation with Chicago's previous day, 0.25 same day) and traced it to a one-day timestamp error; with it fixed, a rule set in advance had no edge out of sample (2010–23)

Delta-Hedging Simulator

MATLAB

- Simulated delta-hedging a short call on 50,000 paths, first checking it against theory (hedging four times as often halved the error, as $1/\sqrt{N}$ predicts), then found 0.1–0.5% costs push the best frequency from daily to weekly
- Reasoned that a timetable wastes trades when the hedge has barely drifted, so tested a no-trade band around delta: on fresh paths the best band cut RMS P&L by 6% vs the best timetable at 0.1% costs, and 2% at 0.5%

Monte Carlo Option Pricing Engine

MATLAB

- Priced an arithmetic Asian call by Monte Carlo, validated against two closed-form prices; chose a geometric-Asian control variate because the two payoffs move together ($\rho = 0.9996$), cutting the variance 1,276-fold
- Mapped where the control variate weakens over 8 volatilities and 6 strikes, and why: at the money its gain falls roughly as $1/\sigma^2$, from 4,569x to 71x as volatility rises from 10% to 80%, and to 52x at worst out of the money

Empirical CAPM Test

MATLAB

- Tested the CAPM on 694 anomaly portfolios in 54 countries (JKP data): 215 alphas (31%) were significant at 5%, over six times the 35 luck alone would produce; GRS rejected it in 48 of 53 countries
- Corrected for running 694 tests at once: 123 survive Benjamini-Hochberg, 84 clear $|t| > 3$ and 22 Bonferroni, nearly all positive where flukes would split by sign; momentum holds up best, size looks like luck

WORK EXPERIENCE

Le Set, Glasgow | Head Mixologist

Sep 2026 – Present

- Designed a new 200-cover venue's 11-drink menu, including 4 original cocktails; trained around 15 bar staff

KONG, Glasgow | Mixologist and Front of House

Oct 2024 – Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%

LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 – Present

- Co-founded Strathclyde's first agri-finance society, grew it to 126 members and published weekly research notes

Strathclyde Poker Society | Vice-President, then President

Sep 2025 – Apr 2026

- Ran weekly tournaments for about 60 players on a £600 budget per semester, reconciling buy-ins and prizes
- Having played 12 tables at once myself, taught new members to use the Kelly criterion to estimate how many buy-ins their win rate, variance and risk level call for, with a buffer in case the edge is overestimated

ADDITIONAL INFORMATION

Technical: R, MATLAB, Excel, Git

Chess: 14th at the 2023 Irish U19 Championships; gave a 22-board simul (average 1860 FIDE) and won 21 games

Powerlifting: Northern Ireland U19 champion (2023)